Months 10 to 12: Fine-tuning

Senior Data Scientist roadmap

B26-L1 Transfer learning and loading pretrained weights

Load only the backbone, use strict=False for the new head, and check missing_keys and unexpected_keys every time.

```
torch.save(model.state_dict(), "encoder.pt")
state = torch.load("encoder.pt")      # weights_only=True by default since 2.6
result = model.load_state_dict(state, strict=False)
result.missing_keys, result.unexpected_keys
```

state_dict	A dictionary from parameter names to tensors. torch.save writes it, torch.load reads it (weights_only=True by default since PyTorch 2.6).
load_state_dict(strict=False)	Loads matching names and returns missing_keys and unexpected_keys. Nothing raises if no name matches, so always check the result.
Shape mismatch	Raises even with strict=False. Drop the old head's keys before loading into a model with a new head.
Expected missing keys	Only the new head. Any other missing key means the backbone did not load.

B26-L2 Learning rates for fine-tuning, warm-up and decay

Warm up from 0, decay to 0, count steps as batches times epochs, and step the scheduler after the optimiser every batch.

```
def warmup_linear(step, warmup_steps, total_steps):
    if step < warmup_steps:
        return step / warmup_steps
    return max(0.0, (total_steps - step) / (total_steps - warmup_steps))
```

Warm-up	Raise the rate from 0 in a straight line over the first steps, so early noisy gradients do not damage pretrained weights.
Linear decay	From the end of warm-up, lower the rate in a straight line to 0 at the last step: (total - step) / (total - warmup).
Steps	Batches per epoch, rounded up, times epochs. Call scheduler.step() after optimizer.step() once per batch.
transformers 5.22	get_linear_schedule_with_warmup(optimizer, num_warmup_steps, num_training_steps); TrainingArguments learning_rate 5e-5, warmup_steps a count or a share below 1.

B26-L3 LoRA, fine-tuning with a low-rank update

LoRA freezes W and trains A and B with B at zero, scales the update by alpha / r, and merges into one matrix after training.

```
y = x @ W.T + (alpha / r) * (x @ A.T @ B.T)
```

LoRA output	y = x @ W.T + (alpha / r) * x @ A.T @ B.T, with W frozen, A (r, d_in), B (d_out, r).
LoRA starting point	A random, B zero, so the update is zero and training starts from the pretrained model.

LoRA size	r * (d_in + d_out) values per layer: 12,288 instead of 589,824 for 768 by 768 at r = 8.
Merging	W + (alpha / r) * B @ A as a new matrix: one plain layer, no extra cost at prediction.

B26-L4 Fine-tuning a text classifier with Hugging Face

Pass num_labels and the label maps, read the load report, use integer labels, call model.train(), and give LoraConfig a task_type.

```
tokenizer = AutoTokenizer.from_pretrained(name)
model = AutoModelForSequenceClassification.from_pretrained(
    name, num_labels=3, id2label=id2label, label2id=label2id)
batch = tokenizer(texts, padding=True, truncation=True, max_length=128,
  return_tensors="pt")
out = model(**batch, labels=labels)       # out.loss, out.logits
```

```
config = LoraConfig(task_type=TaskType.SEQ_CLS, r=8, lora_alpha=16,
  target_modules=["query", "value"])
model = get_peft_model(model, config)
model.print_trainable_parameters()
```

Loading a classifier (transformers 5.22)	AutoModelForSequenceClassification.from_pretrained(name, num_labels=..., id2label=..., label2id=...). The new head shows as MISSING in the load report.
Labels decide the loss	Integer IDs: cross-entropy. Floats with num_labels > 1: multi-label sigmoid. num_labels=1: mean squared error.
Tokenizer call	tokenizer(texts, padding=True, truncation=True, max_length=..., return_tensors="pt") gives input_ids and attention_mask.
peft LoRA for classification (0.21)	LoraConfig(task_type=TaskType.SEQ_CLS, target_modules=["query", "value"], ...). Without task_type the new head stays frozen.

B26-L5 Token classification, named entities and subword labels

Tokenise with is_split_into_words=True, align with word_ids(), label the first piece of each word, and give everything else -100.

```
enc = tokenizer(["Ana", "visited", "London", "staffs"], is_split_into_words=True)
enc.word_ids()     # [None, 0, 1, 2, 3, 3, None]
```

BIO tags	B- begins an entity, I- continues it, O is outside. "New York" is B-LOC I-LOC.
word_ids() (transformers 5.22)	With is_split_into_words=True, the word index of each token: None for special tokens, repeats for split words.
Label alignment	None gets -100, a word's first token gets its label, later pieces get -100. Test with `is None`, never truthiness.
-100	The default ignore_index of nn.CrossEntropyLoss (PyTorch 2.15). Those positions add nothing to the loss.

B26-L6 Evaluating a fine-tuned model

Compare against a trivial baseline and a simple model, report mean and range over seeds, score the test set once, and use entity-level F1 for NER.

```
["B-PER", "O", "B-LOC", "I-LOC"]    # {("PER", 0, 1), ("LOC", 2, 4)}
```

Baselines for a fine-tuned model	The trivial baseline (majority class, or all O), a simple model, and the other fine-tuning strategies, on the same validation set.
Seeds	Run at least three, report mean and range. The best seed reports luck.
Entity-level F1	Spans (type, start, end) must match exactly. Precision over predicted spans, recall over true spans.
Test set	Score it once, at the end. Every extra look turns it into validation data.